

JONEIL CAOILE

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PROFESSIONAL SUMMARY

M.S. Biomedical Engineering candidate (Stevens, 3.9 GPA) with a B.S. in Kinesiology and hands-on experience building device prototypes from concept through bench testing. Background in physiological signal processing, simulated-use design, and CAD-to-3D-print prototype iteration applied to medical device R&D.

TECHNICAL SKILLS

Prototyping & Design: SolidWorks, 3D printing (Bambu Lab), microcontroller-based sensor integration, signal processing, design controls (coursework), risk analysis (ISO 14971 coursework), simulated-use workflow design, protocol development.

Instruments & Signals: EMG signal analysis, motion capture (OptiTrack/Motive), force plates, EKG analysis, auditory feedback (Max/MSP), Neubie electrical stimulation device.

Technical: Python (NumPy, SciPy, Matplotlib), MATLAB, R, SQL (SQLite, PostgreSQL), Excel, Git; biomechanics, cardiopulmonary physiology, neuromuscular physiology.

EDUCATION

Stevens Institute of Technology, *M.S. Biomedical Engineering* Expected May 2026 | GPA 3.9
Concentration: Medical Devices & Clinical Development | Provost Master's Scholarship

Relevant Coursework: Machine Learning, Clinical Research and Design, Biofeedback, Physiology, BMI, Biostatistics

California State University, Long Beach, *B.S. Kinesiology: Exercise Science* Dec 2020 | Cum Laude

PROJECTS

Chest Drain Air Leak Monitor – Inline Retrofit Cartridge (Advisor: Dr. Jinho Kim) Jan 2026 – May 2026 (Expected)

- Designed an inline retrofit cartridge with an integrated pressure tap (MPRLS + FS3000, dual I2C) that adds digital air leak classification to existing chest drains; thresholds per Cerfolio drainology.
- Designed a custom SolidWorks tubing adapter for FS3000 airflow sensor integration; 3D-printed on Bambu Lab with press-fit cavity sized from sensor datasheet specs.
- Defining sensor accuracy acceptance criteria; resolved I2C bus contention with 4.7 kΩ series resistors.

Parkinson's Tremor-Suppression System (Product & Prototyping Lead) Sep 2025 – Dec 2025

- Built a Python (NumPy, SciPy) simulation pipeline (4-8 Hz Butterworth bandpass, RMS envelope, proportional closed-loop decoding) to verify EMG-based tremor detection before hardware implementation.
- Achieved ~45% reduction in simulated tremor amplitude (Pearson $r = 0.994$, RMSE ≈ 0.12); designed wearable EMG cuff concept and self-taught signal processing pipeline from scratch.

Clinician-Facing Reliability Tool: Fall-Risk Assessment (Advisor: Dr. Antonia Zaferiou) Jan 2025 – May 2025

- Built a SQLite-backed tool that cleaned a 77-patient clinical survey dataset and computed inter-rater reliability metrics (Krippendorff's Alpha, percent agreement) for a proposed fall-risk study.
- Designed a clinician-facing dashboard and iterated outputs with faculty stakeholders; designed the SQLite schema for migration to PostgreSQL.

EXPERIENCE

Graduate Researcher Jan 2025 – Present

Stevens Institute of Technology

- Design test protocols and instrumentation for a retrofit chest drain sensor module capstone in Dr. Jinho Kim's lab (Jan 2026–Present).
- Previously: Dr. Antonia Zaferiou's Musculoskeletal Control and Dynamics Lab (Jan-May 2025); see project work above.

Physical Therapy Aide 2016 – 2023 | 400+ Hours

Various Physical Therapy Clinics (Orthopedic, Pediatric, Geriatric)

- Assisted therapists with patient sessions, device setup, and treatment workflow across three clinical settings. Built first-hand understanding of end-user needs for medical device design and testing.

CERTIFICATIONS

Microsoft / LinkedIn Career Essentials (Apr 2026): Data Analysis, Cybersecurity (95%), Business Analysis (95%), Project Management; Atlassian Agile Project Management (94%); Aha! Product Management Certified.